

# Vortex transport and screening in a closed two-dimensional Bose field

Friction, transverse force, diffusion and the finite-wavevector dielectric relation of quantized vortices, measured in an energy-conserving classical field with no reservoir and no fit parameter — pre-registered, with the instrument failures that preceded the measurements

Xavier Callens

SocrateAI-Scientific-QuantumFluids

<https://github.com/xaviercallens/SocrateAI-Scientific-QuantumFluids>

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## Abstract

A quantized vortex in a thermal two-dimensional Bose gas is dragged by the normal fluid (friction  $\alpha$ ), pushed sideways by it (transverse coefficient  $\alpha'$ ) and kicked by it (diffusion  $\eta$ ). The only first-principles values of  $\alpha$  for a bosonic classical field stop at  $0.18 T_{\text{BKT}}$  and carry no error bar on  $\alpha'$  [1]; the fluctuation–dissipation relation that ties  $\eta$  to  $\alpha$  has been proposed for test [2] and found violated by two orders of magnitude in one experiment [3], where walls and noise are suspected. We measure the three coefficients in the projected Gross–Pitaevskii equation — microcanonical, no damping term, no noise put in by hand — with vortex positions tracked to sub-grid precision and estimators that are identities of the dissipative point-vortex model, machine-checked in Lean 4 (`DissipativeVortexDynamics.lean`, `EinsteinRelation.lean`). Every hypothesis, gate and decision rule was committed before the corresponding run. The instrument failed its  $T = 0$  control twice before it passed: a vortex imprint that is not periodic on the torus and an amplitude factor whose sound moves vortices at  $T = 0$  as much as thermal friction does at  $0.14 T_{\text{BKT}}$ ; both are reported, and one earlier “known-answer pass” built on that instrument is withdrawn. With the corrected instrument, at  $T = 0$  a pair neither shrinks nor diffuses over 400 time units and moves at the torus point-vortex velocity to 0.14%; at  $T/T_{\text{BKT}} = 0.14$ ,  $\alpha = 0.0062 \pm 0.0004$  from the energy decay and  $0.0065 \pm 0.0004$  from the trajectory regression, with no dependence on pair size, and  $\alpha' = -0.010 \pm 0.002$ . The warmer temperatures, the Einstein-relation verdict and the dielectric relation on fully equilibrated  $L = 192$  fields are [pending: production analysis and the last three  $L = 192$  runs]. On the way we identify the single non-equilibrium vortex pair that corrupts the superfluid-stiffness estimator in finite boxes (the finite- $k$  form of the Vallat–Beck polarisation term [4, 5]), show that it decays by a factor 3–28 when the box is given  $L^z$ -scaled time ( $z \approx 1.7$ –2 [6, 7]), and find that the Josephson offset  $\eta_{g_1} K - 1$  of this classical field does not go away with box size or geometry:  $+0.31 \pm 0.11$  at  $L = 192$  on equilibrated states. No physics is claimed as new; the numbers are.

## 1 What is measured and why it is open

With the normal fluid at rest a vortex of charge  $q$  moves at

$$\mathbf{v} = (1 - \alpha') \mathbf{v}_s - \alpha q \hat{z} \times \mathbf{v}_s, \quad (1)$$

$\mathbf{v}_s$  the superfluid velocity induced at its position by the other vortices [8, 9]. For  $\alpha \rightarrow 0$  the two sides of a long controversy predict different  $\alpha'$ : no transverse force from the normal fluid,

$\alpha' = 0$  [10], or the Iordanskii force,  $\alpha' = \rho_n/\rho$  [8, 9]. If the field is the vortex’s heat bath, friction and diffusion obey  $\eta = \alpha T/(2\pi\rho)$  ( $\hbar = m = k_B = 1$ ) [2]. For a bosonic classical field,  $\alpha(T)$  exists only below  $0.18 T_{\text{BKT}}$  [1],  $\alpha'$  changes sign within its noise there and “deviates” in 3D Kelvin-wave fits [11]; the Einstein relation has never been tested in a closed field; helium-film analyses assume it [12]; the one cold-atom measurement of  $\eta$  [3] is  $100\times$  the prediction. A review of the field is [13].

A second question is static. The superfluid stiffness is read from the transverse current response,  $\rho_n(k) = \langle |J_T(k)|^2 \rangle / (TL^2)$ ; in the Coulomb-gas picture vortices reduce it through their polarisation [4, 5, 14]. For a compressible Bose field no functional form at finite  $k$  exists: classical-field work extrapolates to  $k \rightarrow 0$  by fits of convenience and has reported negative superfluid fractions attributed to noise [15, 16].

**Model and assets.** Projected Gross–Pitaevskii equation ( $\hbar = m = g = 1$ ,  $dx = \xi/2$ , cutoff at half the grid wavenumber), energy conserved to  $10^{-7}$ – $10^{-6}$  over every run; boxes  $L = 64$  (transport) and  $L = 192$  (equilibration, dielectric relation); a calibrated two-window thermometer;  $T_{\text{BKT}}(L=64) = 0.821$  from the round-2 ladder; thermal states with  $T/T_{\text{BKT}} = 0.14, 0.27, 0.43$  and no thermal vortex (raw count  $\leq 0.36$ ), and one at  $0.56$  with thermal pairs (report only). Pre-registrations, amendments and gate records are files in the repository (docs/designs/PGPE\_FRICTION\_PREREG.md and amendments A1, A1.1, A1.2; PGPE\_ALPHAPRIME\_PREREG.md; PGPE\_EINSTEIN\_PREREG.md; PGPE\_DIELECTRIC\_PREREG.md with amendments D-A1, D-A2), each committed before the data it governs existed.

## 2 The instrument, and what it took to trust it

**Estimators (identities of the model).** Writing the configuration’s point-vortex Hamiltonian  $H$  (Weiss–McWilliams on the torus [17]), Eq. (1) is a skew part plus  $-\frac{\alpha}{2}\nabla H$ , so  $\dot{H} = -\frac{\alpha}{2}|\nabla H|^2$  for any configuration and any  $\alpha'$ :  $\alpha = -\Delta H/(2\int \sum_i |\mathbf{v}_{s,i}|^2 dt)$  is the *energy estimator*, pairing-independent and blind to  $\alpha'$  (*energy\_dissipation*). The displacement regressed on the two orthogonal predictors  $\int \mathbf{v}_s dt$  and  $-q\int \hat{z}\times\mathbf{v}_s dt$  gives  $1-\alpha'$  and  $\alpha$  separately (*centre\_velocity\_identity*); its residuals’ mean square against lag gives  $\eta$ , quoted only if the scaling exponent lies in  $[0.8, 1.2]$ . For a stochastic pair with drift  $-2\alpha\mathbf{d}/|\mathbf{d}|^2$  and diffusion  $D = 2\eta$ , the Kosterlitz–Thouless pair distribution  $|\mathbf{d}|^{-K}$  carries zero probability current iff  $DK = 2\alpha$  (*zero\_flux\_iff*); the Einstein statistic is  $R_E = \eta K/\alpha$ .

**Gates.** Synthetic Langevin trajectories with known  $(\alpha, \alpha', \eta)$  are recovered to 7%, 0.002 and 25% (G0; its first attempt failed on a generator defect, on file). The  $T = 0$  control (G1) failed on its first run and exposed two defects of the imprint used by this programme since round 2: the theta-function phase is periodic only when  $\sum \mathbf{q}\mathbf{r} \in L\mathbb{Z}^2$  — a single pair carries a phase step  $2\pi d/L$  along a whole line — and the amplitude factor  $[r^2/(r^2 + 2)]^{1/2}$  per vortex has a  $1/r^2$  tail four times the Bernoulli depletion, whose sound drifts pair separations by 0.6–0.8 in 400 time units at  $T = 0$ : an apparent  $|\alpha|$  of 0.007–0.010, as large as the thermal value at  $0.14 T_{\text{BKT}}$  (cf. [18] on product-ansatz dipoles). With a periodic phase and a Bernoulli amplitude (*imprint\_v2*) G1 passes:  $|\hat{\alpha}| = 4 \times 10^{-4}$ ,  $1 - \hat{\alpha}' = 0.9986$ , no diffusion, single-pair field momentum constant to  $3 \times 10^{-10}$  and 0.96–0.99 of  $2\pi nd$ , separation constant at 9.82 — a pair at  $T = 0$  neither shrinks nor diffuses [19]. The thermal known answer (G2,  $T/T_{\text{BKT}} = 0.14$ , two antiparallel pairs so that no net impulse is shed) gives  $\alpha = 0.00705 \pm 0.00012$  inside the window set from [1] before the run.

**Withdrawn.** An earlier known-answer gate of this study (single dipoles, the old imprint, a coarse-grained detector biased low by  $\approx 1$  in separation) was reported as “PASS, narrowly” with an unexpected dependence of  $\alpha$  on pair size and a stalled pair. All three were artefacts of that instrument and are withdrawn (LEDGER CLAIM-078). The stall motivated a hypothesis — the impulse shed by a shrinking pair in a closed box drives a phonon wind that stalls it (`wind_stall`) — which stays registered; its status is in Section 4.

### 3 Results at $T/T_{\text{BKT}} = 0.14$ (complete) and warmer (pending)

Eight runs (antiparallel pairs,  $d_0 = 8$  and 12, 2000 time units, corrected imprint, raw detection with sub-grid refinement):

|                                    | $d_0 = 8$           | $d_0 = 12$          | all                                                            |
|------------------------------------|---------------------|---------------------|----------------------------------------------------------------|
| $\alpha$ , energy estimator        | $0.0057 \pm 0.0006$ | $0.0063 \pm 0.0011$ | <b><math>0.0062 \pm 0.0004</math></b>                          |
| $\alpha$ , regression              | $0.0059 \pm 0.0005$ | $0.0063 \pm 0.0010$ | $0.0065 \pm 0.0004$                                            |
| $\alpha'$                          | $-0.014 \pm 0.002$  | $-0.004 \pm 0.002$  | <b><math>-0.010 \pm 0.002</math></b>                           |
| $\eta$ (not quoted: exponent 0.71) | —                   | —                   | $(3.7 \times 10^{-4}$ if read; Einstein $1.2 \times 10^{-4}$ ) |

Friction is  $17\times$  the  $T = 0$  floor, within a factor two of [1] interpolated to this temperature and of an independent configuration of this programme (eight imprinted vortices,  $\alpha = 0.0056$ ), and shows no pair-size dependence (criterion F1': ratio 0.91, window [0.7, 1.43]). The transverse coefficient is small and *negative*: vortices move 1% faster than the torus point-vortex velocity. Two facts bear on its reading before any Iordanskii-versus-no-force statement:  $\alpha'$  differs between the two pair sizes by five standard errors, and at  $T = 0$  a pair of separation 3–6 already moves 5% faster than the point-vortex prediction (compressible correction to the pair speed; measured here on 60-time-unit runs, credible only for tight pairs). The registered test regresses  $\alpha'$  on  $\rho_n/\rho$  over three temperatures; the  $T = 0$  baseline must be subtracted size by size. Residual motion is sub-diffusive on these time scales (exponent 0.71), so by the registered rule  $\eta$  is not quoted at this temperature.

**Warmer temperatures.** The first production pass at  $T/T_{\text{BKT}} = 0.27, 0.43, 0.56$  produced no data: thermal jitter of the detected core exceeds the 1.5 tracking radius (per-sample displacement 95th percentile 0.98–1.0 against 0.60 at 0.14), and every track was lost inside the settling window (amendment A1.2, radius 3.0, nothing else changed). The repeat finished with usable tracks (556–2000 time units at 0.27, three natural annihilations; 118–729 at 0.43). [pending: registered F2, F3, the  $\alpha'$  regression and  $R_E$  — to be computed by the rules as written].

### 4 The phonon wind of a closed box

A pair of separation  $d$  carries impulse  $2\pi\rho_s d$ ; in a periodic box its decrease has nowhere to go but the phonons, which then drift at  $u = 2\pi\rho_s(d_0 - d)/(\rho_n L^2)$  and reduce the friction drive:  $\dot{d} = -2\alpha(1/d - u)$ , which stalls at  $b = [d_0 + (d_0^2 - 2\rho_n L^2/\pi\rho_s)^{1/2}]/2$  when the root is real (`wind_stall`, machine-checked). At  $T/T_{\text{BKT}} = 0.14$ ,  $L = 64$ :  $d_0 = 12$  should stall (predicted  $b = 10.2$ ),  $d_0 = 8$  should not. Two single-pair runs with the corrected imprint, 4000 time units: both pairs stop shrinking — one at 9.3 after 2000 units, one re-expands from 10.2 to 11.2 — while antiparallel pairs at the same temperature shrink steadily from 10 to 7–8; the momentum in the phonon band grows with the impulse the pair loses, slopes 0.68 and 0.52 (registered window [0.5, 1.2]). The plateau is below the predicted 10.2. The decisive test, the same pair in an  $L = 96$  box where no stall is predicted, has not been run. If the mechanism holds, closed-box friction

measurements on single pairs at low temperature are contaminated unless  $\rho_n L^2 \gg \pi \rho_s d_0^2/2$  — a condition the literature’s truncated-GPE measurements should be checked against.

## 5 Box-scale vortex charge, equilibration and the stiffness estimator

Seven of eighteen large-box runs of an earlier round reached  $n_s/n < 0$  with the same vortex number as healthy partners. A pre-registered test on their vortex positions refuted the Onsager-cluster reading (no like-sign clustering, positive vortex temperature, 98–99.7% of vortices paired with an opposite neighbour); post hoc, the point-vortex transverse current  $n_{\text{eff}}^2 (2\pi)^2 |\rho_q(k)|^2/k^2$  computed from positions alone tracks the measured one across all eighteen runs with  $r = 0.996$ : the anomaly is one  $O(1)$  unscreened pair at the box scale — the finite- $k$  form of the polarisation term of [4, 5], the single free vortex of early superconducting films, and the post-quench surviving pair that Chu and Williams asked simulations to look for [20]. Such a pair has Boltzmann weight  $(L/\xi)^{-K} \lesssim 10^{-9}$ : it is a remnant of the initial relaxation. Equilibrium finite-size stiffness is monotone in  $L$  [21, 22]; coarsening under conservative dynamics gives  $z \approx 1.7$ – $1.8$  [6] ( $L^2 \ln L$  for dissipative XY [7, 23];  $L^2/\ln L$  Hamiltonian XY [24]), so the warm-up that cured  $L = 128$  should scale to  $\approx 11\,000$ – $14\,600$  at  $L = 192$ . Registered before the runs: continuing the six  $L = 192$  trajectories to  $t = 14\,500$  should admit at least five of six, bring the vortex transverse current below 0.45 in every admitted run, and drop it by at least  $2\times$  in the two anomalous ones. Three of six are in:

| run                       | admitted | $R_T^{\text{vortex}}: t \in [6000, 7000] \rightarrow [13500, 14500]$ | $n_s/n$                   | $\eta_{g_1} K - 1$ |
|---------------------------|----------|----------------------------------------------------------------------|---------------------------|--------------------|
| e1.00 s11 (was anomalous) | yes      | $1.98 \rightarrow 0.07$                                              | $-0.38 \rightarrow +0.81$ | $+0.26$            |
| e1.00 s12                 | yes      | $0.65 \rightarrow 0.08$                                              | $+0.49 \rightarrow +0.77$ | $+0.13$            |
| e1.10 s12                 | yes      | $0.74 \rightarrow 0.19$                                              | $+0.41 \rightarrow +0.80$ | $+0.51$            |

[pending: the other three runs, including the second anomalous one; verdict by the registered rule].

**Josephson relation.** On these equilibrated  $L = 192$  states the coherence exponent fitted with the exact torus Green function on  $r \in [8, 32]$  gives  $\eta_{g_1} K - 1 = +0.31 \pm 0.11$ , unchanged by the fit window (16–48) and equal to the naive-window value: the 12–27% excess reported at  $L \leq 64$  is neither a geometry factor (cf. “close to unity” [25]) nor a finite-size transient. It persists in equilibrium at  $L = 192$ . What remains is the estimator pair itself —  $\eta$  from  $g_1$  against  $K$  from current correlators on a fixed-momentum microcanonical field, for which no fluctuation formula has been derived — or the renormalisation of the bare stiffness by amplitude fluctuations. Reported, not explained; the earlier statement that this excess is “unexplained finite-size” is amended.

## 6 The finite-wavevector dielectric relation

Pre-registered claims: mode by mode and snapshot by snapshot,  $J_T(k) = n_{\text{eff}} (2\pi i \rho_q(k)/|k|)$  plus a phonon part uncorrelated with  $\rho_q$  (coherence  $\gamma^2 \geq 0.4$  on every shell  $|m|^2 \leq 16$  in  $\geq 5$  of 6 runs), with one real  $k$ -independent  $n_{\text{eff}}$ . Five gate runs on imprinted pair gases failed as written and are on file; the coherence criterion passed in all five ( $\gamma^2 = 1.000$  at  $T = 0$  with the corrected imprint, residual power  $10^{-4}$ ): the relation exists mode by mode. What moved was the amplitude’s known answer: it is not 1 for pairs of finite size but rises with pair size, 0.922, 0.971, 0.991 at  $d = 5, 8, 14$  (density is depleted where the flow is fast, so a tight pair carries less current than two point vortices); a static Bernoulli form factor is low by 0.02–0.04. For pairs

of polarisation form  $\rho_q(k) = i\mathbf{k} \cdot \mathbf{P}(k)$  with explicit remainder, and the  $k$ -independent ceiling of the bound-pair response, see `PairPolarisation.lean`. [pending: D1, D2 on the six equilibrated  $L = 192$  runs;  $n_{\text{eff}}$  against the three registered closures with no favourite]. The size dependence has a consequence worth stating now: thermal pairs sit at  $d \approx 1\text{--}3$ , where the amplitude is  $\lesssim 0.9$ , so part of the stiffness reduction attributed to vortices near  $T_{\text{BKT}}$  is compressible, not Coulomb-gas.

## 7 What this establishes, and what it does not

### Established.

- At  $T = 0$  a vortex pair in the projected GPE neither shrinks nor diffuses over 400 time units and moves at the torus point-vortex velocity to 0.14%; tight pairs ( $d \leq 6$ ) move 5% faster.
- At  $T/T_{\text{BKT}} = 0.14$ :  $\alpha = 0.0062 \pm 0.0004$  by two estimators that agree, no pair-size dependence,  $\alpha' = -0.010 \pm 0.002$ ; the fluctuation of the separation is sub-diffusive on 2000 time units.
- The depressed-stiffness states of finite boxes are not vortex clusters; they carry one un-screened box-scale pair whose transverse current accounts for the measured one ( $r = 0.996$ ); given  $L^z$ -scaled time it decays by  $3\text{--}28\times$  (three of six runs).
- The Josephson offset  $\eta_{g1}K - 1 = +0.31 \pm 0.11$  survives at  $L = 192$  in equilibrium, independent of window and torus geometry.
- The transverse current of a pair gas is the point-vortex term mode by mode (coherence 1.000 at  $T = 0$ , 0.99 at  $0.14 T_{\text{BKT}}$ ), with an amplitude that depends on pair size.
- Theorems: energy dissipation of any skew-plus-gradient vortex flow, the  $d^2$  law and its lifetime bound, the centre-velocity identity, the wind stall, the zero-flux/Einstein equivalence, the pair-polarisation form and response ceiling, the charge-density/matching bound and the Kosterlitz-flow invariant (all standard axioms).

### Not established.

- $\alpha(T)$  above  $0.14 T_{\text{BKT}}$ , the  $\alpha'$  regression, and the Einstein relation: the data exist, the registered analysis has not been run.
- The phonon-wind mechanism: two single pairs stall as predicted in direction, not in value; the  $L = 96$  control does not exist.
- The cause of the Josephson offset; the quantitative form factor of a pair's current.
- Whether  $\alpha' < 0$  is a normal-fluid force or a compressible correction to the pair speed: separable only after the  $T = 0$  baseline is measured over 100–400 time units at each size.
- Anything about quantum gases or helium: this is a Rayleigh–Jeans field with one cutoff,  $mg = 1$ , two dimensions.

**Process record.** Three gate failures on the transport instrument (one generator, two imprint), one withdrawn gate verdict, one tracking amendment after a production pass with no data; five dielectric gate failures on the amplitude's known answer with the coherence criterion untouched; every hypothesis ranked by a five-minute probe under priors fixed before the probe, with the

ranking's sensitivity reported. Data, code, pre-registrations and the Lean sources are in the repository; the literature base behind the design (773 entries, 382 arXiv identifiers verified against the arXiv API) is described in `docs/designs/LITERATURE_REVIEW_2026-10.md`.

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